AAE 33401 Aerodynamics Laboratory
1 credit

1 Course Information/Description

A laboratory course designed to illustrate various aerodynamic phenomena. Airfoil pressure distribution. Finite wing lift, drag, and moment. Supersonic flow around bodies. Shock waves.

This course is designed to reinforce concepts in compressible fluid mechanics and wing aerodynamics taught in AAE 334, laminar airfoils, flaps and slats, pressure on an airfoil, finite wings, wing tip vortices and wing tip devices, supersonic wind tunnel, normal and oblique shock waves, expansion fans, compressible nozzle flow, optical flow visualization, wind tunnel techniques and instrumentation.**Learning Resources and Reference Text**

Brightspace Learning Management System:

Access the course videos, materials, and resources via Purdue's BS learning management system. Begin with the Start Here tab, which describes how the course Brightspace is organized. We strongly suggest that you explore and become familiar not only with the site navigation but with content and resources available for this course. See the Student Services widget on the campus homepage for resources such as Technology Help, Academic Help, Campus Resources, and Protect Purdue.

Textbook:

There is no required textbook for this course.

2 Assignments

Lab teams will perform five pre-designed laboratories and then design and conduct their own experiment as a final project. The course has individual assignments for the first five labs and the final project will be done as a team. The types of assignments include:

2.1 Individual Pre-Lab Assignments

Complete a short online assignment to prepare for the upcoming lab. The assignments include mostly multiple choice and true/false questions, with occasional file upload required.

2.2 Individual Post-Lab Assignments

Each student is required to individually complete and submit a post-lab assignment after each of the first five labs. The assignment will generally focus on evaluating the objectives of the lab, presentation and analysis of the data, and assessment of error and uncertainty.

2.3 Team Assignment: Team Term Project

Each lab team will be required to think of a novel experiment and submit a short proposal. The team will then set up and complete the experiment, after which they will submit one complete

engineering lab report as a team. Peer evaluations will be used to assess each student's participation in the final project.

3 Course Learning Objectives

- Use wind tunnel instrumentation to measure flow velocity and lift and drag.
- Describe the dependence of airfoil pressure distribution, lift, and drag on angle of attack and flaps and slats.
- Describe the effect of finite span, wing tip vortices.
- Use schlieren or shadowgraph techniques to visualize compressible flows.
- Sketch pressure distributions across oblique and normal shock waves, and expansion fans.
- Describe the pressure distribution inside a compressible flow converging-diverging nozzle and possible nozzle exit conditions.
- Design an experiment to meet a defined objective, conduct the experiment, analyze the data, and draw conclusions.

4 Evaluation

Students are required to attend their bi-weekly labs sessions. Attendance and participation accounts for 20% of the grade for each lab. Late arrival to the lab may result in deduction of half the attendance credit. If you have to miss a lab session due to illness or other extenuating circumstances, please inform the instructor and your TA as early as possible to schedule a make-up time. Obtain the lab data from a team member so you can complete your post-lab assignment.

All assignments must be submitted on Gradescope as instructed. No email submissions will be accepted. Each student is permitted one 1-day extension on a pre-lab assignment and one 2-day extension on a post-lab assignment without penalty. Extension requests must be submitted before the assignment due date using the form in Brightspace. Late pre-lab assignments without an approved extension request will not be accepted. Late post-lab assignments without an approved extension will incur a 5-point per day penalty.

Your grade will be calculated as follows:

- Labs 1-5: 75%
  - Each lab count is weighted equally (i.e., 15% of the final grade)
  - Each lab includes a pre-lab assignment (20%), attendance and participation (20%), and a post-lab assignment (60%).
- Final Project: 25%

Your course grade will be based on the following grading scale:

Letter Grade	Percentage
A+	98-100%
A	93-97.99%
A-	90-92.99%
B+	87-89.99%
B	83-86.99%
B-	80-82.99%
C+	77-79.99%
C	73-76.99%
C-	70-72.99%

Letter Grade	Percentage
D+	67-69.99%
D	63-66.99%
D-	60-62.99%
F	0-59.99%